

Chapter 2

Mathematical preliminaries

In this chapter, we introduce the essential mathematical tools and methodologies that will be used for our analysis in the subsequent sections of the thesis. Initially, we provide a brief general introduction of EOPs. Later we discuss individually X_1 - and X_m -EOPs and their properties.

Further, we discuss various techniques to solve Schrödinger equation, namely SUSY QM and SI, PCT approach, and group theoretic approach.

2.1 Exceptional orthogonal polynomials

There is a broad class of polynomials known as Heine-Stieltjes polynomials $y(x)$ [235, 236], which satisfy a second-order differential equation

$$p(x)y'' + q(x)y' + r(x)y = 0, \quad (2.1)$$

where m is a positive integer and $p(x)$ and $q(x)$ are polynomials with $\deg p = m + 2$ and $\deg q = m + 1$ respectively and $r(x)$ represents an unknown polynomial of degree m , which can depend on the solution $y(x)$. Heine showed that for given an integer n , there are at most $(n + m)!/m!n!$ distinct polynomials $r(x)$ for which $y(x)$ has a polynomial solution of degree n . The $m = 0$ case represents a significant subcategory within the Heine-Stieltjes problem and Eq. (2.1) represents a form of the hypergeometric equation, which yields COPs as its solutions. Bochner [237] transformed Eq. (2.1) into an eigenvalue equation

$$Oy = \lambda_n y, \quad (2.2)$$

with a second-order differential operator

$$Oy = p(x)y'' + q(x)y' + r(x)y. \quad (2.3)$$

If Eq (2.2) has a polynomial solution $y = P_n(x)$ of degree n , where $n = 0, 1, 2, \dots$. Then this must necessarily be of the Heine-Stieltjes type with $m = 0$ and coefficients $p(x)$, $q(x)$ and $r(x)$ are polynomials in x with degree 2, 1 and 0 respectively. The sequence of polynomials $\{P_n(x)\}_{n \geq 0}$ is the system of the COPs such as Laguerre, Jacobi, or Hermite as proved by Lesky [238].

David Gomez-Ullate, Niky Kamran, and Robert Milson [160, 162, 163, 228, 239] modify the differential operator in Eq. (2.3) by redefining the coefficient functions $p(x)$, $q(x)$, and $r(x)$ in the rational form as follows

$$p(x) = \frac{\tilde{p}(x)}{s(x)}, \quad q(x) = \frac{\tilde{q}(x)}{s(x)}, \quad \tilde{r}(x) = \frac{\tilde{r}(x)}{s(x)}, \quad (2.4)$$

where $\tilde{p}(x)$, $\tilde{q}(x)$, $\tilde{r}(x)$, and $s(x)$ are polynomials. After clearing the denominators, the eigenvalue Eq. (2.2) becomes a special form of the Heine-Stieltjes type Eq. (2.1), given as

$$\tilde{p}(x)y'' + \tilde{q}(x)y' + (\tilde{r}(x) - \lambda_n s(x))y = 0. \quad (2.5)$$

They solved above Eq. (2.5) by requiring that the polynomial sequence $\{P_n(x)\}_{n=m}^{\infty}$ commence with a polynomial of degree $m \geq 1$, instead of initiating with a constant P_0 . They impose the condition that the sequence of these polynomials is complete with respect to a positive weight function and discover a new family of orthogonal polynomials termed EOPs. Initially, they explicitly present the case of X_1 -EOPs [228, 239] for $m = 1$ and then generalized these results for any arbitrary m values (X_m -EOPs) [160, 163].

2.2 X_1 -Exceptional Orthogonal Polynomials

For X_1 -EOPs $m = 1$. Let

$$p(x) = k_2(x - b)^2 + k_1(x - b) + k_0, \quad (2.6)$$

where b and c are two distinct constant and k_0 satisfies $k_0 = p(b) \neq 0$. By setting

$$a = \frac{1}{(b-c)}, \quad (2.7)$$

$$\tilde{q}(x) = a(x-c)(k_1(x-b) + 2k_0), \quad (2.8)$$

$$\tilde{r}(x) = -a(k_1(x-b) + 2k_0), \quad (2.9)$$

and defining a second-order differential operator

$$Oy = p(x)y'' + \frac{\tilde{q}(x)}{x-b}y' + \frac{\tilde{r}(x)}{x-b}y, \quad (2.10)$$

one observes that the eigenvalue equation (2.2) with above operator O , is equivalent to an $m = 1$, Heine-Stieltjes type equation

$$(x-b)p(x)y'' + \tilde{q}(x)y' + (\tilde{r}(x) - \lambda(x-b))y = 0, \quad (2.11)$$

where k_2, k_1, k_0, a, b and c are constant parameters. Using the above discussion, Ullate et al. extended Bochner's results to the family of EOPs and found X_1 -Laguerre and X_1 -Jacobi EOPs. The X_1 subspace for the differential operator O is given by $a(x-c), (x-b)^2, (x-c)^2, \dots, (x-b)^n$. We have

$$O(x-c) = 0, \quad (2.12)$$

$$O(x-b)^2 = (2k_2 + ak_1)(x-b)^2 + 2k_0a(x-c), \quad (2.13)$$

$$\begin{aligned} O(x-b)^n &= (n-1)(nk_2 + ak_1)(x-b)^n \\ &\quad + (n(n-2)k_1 + 2(n-1)ak_0)(x-b)^{n-1} \\ &\quad + n(n-3)k_0(x-b)^{n-2}, \quad n \geq 2, \end{aligned} \quad (2.14)$$

from Eq. (2.14) the eigenvalues λ_n are written by

$$\lambda_n = (n-1)(nk_2 + ak_1), \quad n \geq 1. \quad (2.15)$$

Specific examples of EOPs in this category are written below.

2.2.1 X_1 -Jacobi EOPs

To construct X_1 -Jacobi EOPs [228, 239], one sets

$$p(x) = x^2 - 1, \quad (2.16)$$

and from Eq. (2.6), one of the possible sets of constant parameters is

$$k_2 = 1, \quad k_1 = 2b \quad \text{and} \quad k_0 = b^2 - 1. \quad (2.17)$$

By substituting these values, Eq. (2.11) takes the form

$$(x^2 - 1)y'' + 2a \left(\frac{1 - bx}{b - x} \right) ((x - c)y' - y) = \lambda_n y, \quad (2.18)$$

where $y(x)$ defined on the line $-1 \leq x \leq 1$ and follow the boundary conditions

$$\lim_{x \rightarrow 1^-} (1 - x)^{\alpha+1} (y(x) - (x - c)y'(x)) = 0, \quad (2.19)$$

$$\lim_{x \rightarrow -1^+} (1 + x)^{\beta+1} (y(x) - (x - c)y'(x)) = 0. \quad (2.20)$$

The constants α and β are two distinct real parameters such that $\alpha \neq \beta$ and the parameters a , b and c are defined in terms of α and β as

$$a = \frac{1}{2}(\beta - \alpha), \quad b = \frac{\beta + \alpha}{\beta - \alpha}, \quad (2.21a)$$

$$c = b + \frac{1}{a}. \quad (2.21b)$$

The following constraints will be necessary for the real parameters α and β

$$\alpha > -1, \quad \beta > -1, \quad (2.22a)$$

$$\text{sgn } \alpha = \text{sgn } \beta. \quad (2.22b)$$

which ensures $|b| > 1$. The family of X_1 -type Jacobi polynomials is obtained by orthogonalizing the sequence $(x - c), (x - b)^2, (x - b)^3, \dots, (x - b)^n, \dots$ with positive definite inner product

$$\langle P, Q \rangle_{\alpha, \beta} := \int_{-1}^1 \hat{W}_{\alpha, \beta} P(x) Q(x) dx, \quad (2.23)$$

weight function $\hat{W}_{\alpha, \beta}$ is defined in the interval $-1 \leq x \leq 1$ and related to classical weight function as

$$\hat{W}_{\alpha, \beta} = \frac{W_{\alpha, \beta}}{(x - b)^2}, \quad (2.24)$$

where $W_{\alpha, \beta} = (1 - x)^\alpha (1 + x)^\beta$.

From Eq. (2.18) the X_1 -Jacobi polynomials $y = \hat{P}_n^{(\alpha, \beta)}(x)$ for an integer $n \geq 1$ satisfy the differential equation

$$\begin{aligned} \hat{P}_n^{(\alpha, \beta)}(x) + \left(\frac{(\beta + \alpha + 2)x - (\beta - \alpha)}{x^2 - 1} + \frac{2(\beta - \alpha)}{(\beta + \alpha) - (\beta - \alpha)x} \right) \hat{P}_n^{(\alpha, \beta)}(x) \\ + \left(\frac{(\beta - \alpha)x - (n - 1)(n + \alpha + \beta)}{x^2 - 1} + \frac{(\beta - \alpha)^2}{(\beta + \alpha) - (\beta - \alpha)x} \right) \hat{P}_n^{(\alpha, \beta)}(x) = 0. \end{aligned} \quad (2.25)$$

The inner product for X_1 -Jacobi polynomials is given by

$$\int_{-1}^1 \frac{(1-x)^\alpha (1+x)^\beta}{(x-b)^2} (\hat{P}_n^{(\alpha, \beta)}(x))^2 dx = \frac{(\alpha + n)(\beta + n)}{4(\alpha + n - 1)(\beta + n - 1)} C_{n-1}, \quad (2.26)$$

where constant

$$C_n = \frac{2^{\alpha+\beta+1}}{(\alpha + \beta + 2n + 1)} \frac{\Gamma(\alpha + n + 1)\Gamma(\beta + n + 1)}{\Gamma(n + 1)\Gamma(\alpha + \beta + n + 1)} \quad (2.27)$$

is the normalization constant for classical Jacobi orthogonal polynomials $P_n^{(\alpha, \beta)}(x)$.

The relation between X_1 -Jacobi polynomials $\hat{P}_n^{(\alpha, \beta)}$ and Classical Jacobi polynomials $P_n^{(\alpha, \beta)}(x)$ is given by

$$\hat{P}_n^{(\alpha, \beta)} = -\frac{1}{2}(x - b)P_{n-1}^{(\alpha, \beta)} + \frac{bP_{n-1}^{(\alpha, \beta)} - P_{n-2}^{(\alpha, \beta)}}{(\alpha + \beta + 2n - 2)}. \quad (2.28)$$

The first three X_1 -Jacobi polynomials are given as

$$\begin{aligned} \hat{P}_1^{(\alpha, \beta)}(x) &= -\frac{1}{2}x - \frac{2+\alpha+\beta}{2(\alpha-\beta)}, \\ \hat{P}_2^{(\alpha, \beta)}(x) &= -\frac{\alpha+\beta+2}{4}x^2 - \frac{\alpha^2+\beta^2+2(\alpha+\beta)}{2(\alpha-\beta)}x - \frac{\alpha+\beta+2}{4}, \\ \hat{P}_3^{(\alpha, \beta)}(x) &= -\frac{(\alpha+\beta+3)(\alpha+\beta+4)}{16}x^3 - \frac{(3+\alpha+\beta)(6\alpha+3\alpha^2+6\beta-2\alpha\beta+3\beta^2)}{16(\alpha-\beta)}x^2 \\ &\quad - \frac{(9\alpha+3\alpha^2+9\beta+2\alpha\beta+3\beta^2)}{16}x - \frac{-6\alpha+\alpha^2+\alpha^3-6\beta-6\alpha\beta-\alpha^2\beta+\beta^2-\alpha\beta^2+\beta^3}{16(\alpha-\beta)}. \end{aligned}$$

For other properties related to X_1 -Jacobi polynomials, see Ref. [228, 239].

2.2.2 X_1 -Laguerre polynomials

To construct X_1 -Laguerre polynomials one considers the $p(x)$ by setting the parameters $k_2 = 0$, $k_1 = -1$, $k_0 = k$, $b = -k$ and $a = -1$ in Eq. (2.6).

Substituting these values in Eq. (2.11) and for $k > 0$, one obtains

$$-xy'' + \left(\frac{x-k}{x+k} \right) ((x+k+1)y' - y) = \lambda_n y, \quad (2.30)$$

where $y(x)$ is a twice differentiable function that is defined on the line $0 \leq x \leq \infty$ and restricted by the following boundary conditions

$$\lim_{x \rightarrow 0^+} x^{k+1} e^{-x} (y(x) - (x - c)y'(x)) = 0, \quad (2.31)$$

$$\lim_{x \rightarrow \infty} x^{k+1} e^{-x} (y(x) - (x - c)y'(x)) = 0. \quad (2.32)$$

The X_1 -Laguerre polynomials are obtained by orthogonalizing the sequence $(x + k + 1), (x + k)^2, (x + k)^3, \dots, (x + k)^n, \dots$ with a positive definite inner product

$$\langle P, Q \rangle_k := \int_0^\infty \hat{W}_k P(x) Q(x) dx, \quad (2.33)$$

weight function $\hat{W}_k$ is defined in the interval $0 \leq x \leq \infty$ and related to classical Laguerre weight function as

$$\hat{W}_k = \frac{W_k}{(x + k)^2}, \quad (2.34)$$

where Laguerre weight function $W_k = x^k e^{-x}$.

From Eq. (2.30) the X_1 -Laguerre polynomials $y(x) = \hat{L}_n^k(x)$ with $n = 1, 2, 3, \dots$ satisfy the following differential equation

$$-x \hat{L}_n^{(k)''}(x) + \frac{(x - k)}{x + k} [(x + k + 1) \hat{L}_n^{(k)'}(x)] - \left[\frac{(x - k)}{(x + k)} + (n - 1) \right] \hat{L}_n^{(k)} = 0 \quad (2.35)$$

The inner product for X_1 -Laguerre polynomials is given by

$$\int_0^\infty \frac{x^k e^{-x}}{(x + k)^2} (\hat{L}_n^k(x))^2 dx = \frac{k + n - 1}{k + n} C_{n-1}', \quad (2.36)$$

where constant

$$C_n = \frac{\Gamma(k + n + 1)}{n!}. \quad (2.37)$$

is the normalization constant corresponding to classical Laguerre polynomials $L_n^k(x)$ of degree n

The relation between X_1 -Laguerre polynomials $\hat{L}_n^{(k)}(x)$ and classical Laguerre polynomials $L_n^k(x)$ is given by

$$\hat{L}_n^k = -(x + k + 1) L_{n-1}^{(k)} + L_{n-2}^{(k)}. \quad (2.38)$$

First three X_1 -Laguerre polynomials are given as

$$\hat{L}_1^k(x) = -(x + k + 1), \quad (2.39a)$$

$$\hat{L}_2^k(x) = x^2 - k(k + 2), \quad (2.39b)$$

$$\hat{L}_3^k(x) = -\frac{1}{2}x^3 + \frac{k+3}{2}x^2 + \frac{k(k+3)}{2}x - \frac{k}{2}(3 + 4k + k^2). \quad (2.39c)$$

For other properties related to X_1 -Laguerre polynomials, see the Ref. [228, 239].

2.3 X_m -Exceptional Orthogonal Polynomials

We now turn our attention to a more general case, applicable to any positive integer m , referred to as the X_m orthogonal polynomials. In this section, we briefly discuss X_m -Jacobi and X_m -Laguerre orthogonal polynomials [152, 153, 160, 163].

2.3.1 X_m -Jacobi polynomials

The X_m -Jacobi polynomials for any given positive integer satisfy the differential equation as [163]

$$\begin{aligned} & \hat{P}_{n,m}^{(\alpha,\beta)}(x) + \left[(\alpha - \beta - m + 1) \frac{P_{m-1}^{(-\alpha,\beta)}(x)}{P_m^{(-\alpha-1,\beta-1)}(x)} - \left(\frac{\alpha + 1}{1-x} \right) + \left(\frac{\beta + 1}{1+x} \right) \right] \\ & \times \hat{P}_{n,m}'^{(\alpha,\beta)}(x) + \frac{1}{(1-x^2)} \left[\beta(\alpha - \beta - m + 1)(1-x) \frac{P_{m-1}^{(-\alpha,\beta)}(x)}{P_m^{(-\alpha-1,\beta-1)}(x)} \right. \\ & \left. + m(\alpha - \beta - m + 1) + (n-m)(\alpha + \beta + n - m + 1) \right] \hat{P}_{n,m}^{(\alpha,\beta)}(x) = 0. \end{aligned} \quad (2.40)$$

The X_m -Jacobi polynomials $\hat{P}_{n,m}^{(\alpha,\beta)}(x)$ are formally orthogonal with respect to a weight function $\hat{W}_{\alpha,\beta,m}(x)$ defined in the interval $-1 \leq x \leq 1$, where

$$\hat{W}_m^{\alpha,\beta}(x) = \frac{(1-x)^\alpha(1+x)^\beta}{\left[P_m^{(-\alpha-1,\beta-1)}(x) \right]^2}. \quad (2.41)$$

The L^2 norm corresponding to X_m -Jacobi polynomials is given by

$$\int_{-1}^1 \frac{(1-x)^\alpha(1+x)^\beta}{\left[P_m^{(-\alpha-1,\beta-1)}(x) \right]^2} (\hat{P}_{n,m}^{(\alpha,\beta)}(x))^2 dx = \frac{(1+\alpha+n-2m)(\beta+n)}{(\alpha+n+1-m)^2} C_n^m, \quad (2.42)$$

where constant C_n^m is given by

$$C_n^m = \frac{2^{\alpha+\beta+1} \Gamma(\alpha+2+n-m) \Gamma(\beta+n-m)}{(n-m)! (\alpha+\beta+2n-2m+1) \Gamma(\alpha+\beta+n-m+1)}. \quad (2.43)$$

To achieve L^2 orthogonality, as indicated by equation (2.42), it is essential to impose further conditions on the parameters α, β , and m . The denominator clearly must not equal zero, which results in the following two conditions

$$\begin{aligned} (i) \quad & \beta \neq 0, \quad \alpha, \alpha - \beta - m + 1 \notin \{0, 1, \dots, m-1\}, \\ (ii) \quad & \alpha > m-2, \quad \text{sgn}(\alpha - m + 1) = \text{sgn}(\beta). \end{aligned} \quad (2.44)$$

The relation between exceptional Jacobi polynomials and classical Jacobi polynomials is given by

$$\begin{aligned} \hat{P}_n^{(\alpha, \beta)}(x) = \frac{(-1)^m}{\alpha+1+j} \left(\frac{1}{2} (1+\alpha+\beta+j)(x-1) P_m^{(-\alpha-1, \beta-1)} P_{j-1}^{(\alpha+2, \beta)} \right. \\ \left. + (\alpha+1-m) P_m^{(-\alpha-2, \beta)} P_j^{(\alpha+1, \beta-1)} \right). \end{aligned} \quad (2.45)$$

For a particular value of m , i.e., $m = 0$, we can reconstruct the classical Jacobi weight function and polynomials

$$W_0^{\alpha, \beta} = (1-x)^\alpha (1+x)^\beta, \quad (2.46)$$

$$\hat{P}_{n,0}^{(\alpha, \beta)} = P_n^{(\alpha, \beta)}. \quad (2.47)$$

For $m = 1$, one can obtain the X_1 -Jacobi polynomials and corresponding weight function.

2.4 X_m -Laguerre polynomials

In this section, we provide a brief description of a more general case X_m -Laguerre polynomials for any $m \geq 0$. X_m -Laguerre polynomials $\hat{L}_{n,m}^k$ satisfy the differential equation [160]

$$\begin{aligned} \hat{L}_{n,m}^{''k}(x) + \frac{1}{x} \left((k+1-x) - 2x \frac{L_{m-1}^k(-x)}{L_m^{k-1}(-x)} \right) \hat{L}_{n,m}^{'k}(x) \\ + \frac{1}{x} \left(n - 2k \frac{L_{m-1}^k(-x)}{L_m^{k-1}(-x)} \right) \hat{L}_{n,m}^k(x) = 0. \end{aligned} \quad (2.48)$$

X_m -Laguerre polynomials are orthogonal with respect to a weight function $W_m^k(x)$ defined in the interval $x \in [0, \infty)$, where

$$W_m^k(x) = \frac{x^k e^{-x}}{\left[L_m^{k-1}(-x)\right]^2}. \quad (2.49)$$

For X_m -Laguerre polynomials the L^2 norm is defined as

$$\int_0^\infty \frac{x^k e^{-x}}{\left[L_m^{k-1}(-x)\right]^2} (\hat{L}_{n,m}^{(k)}(x))^2 dx = \frac{(k+n)\Gamma(k+n-m)}{(n-m)!}. \quad (2.50)$$

The X_m polynomials can also be expressed in terms of their classical counterparts. For instance, the n th degree exceptional X_m -Laguerre polynomial is related to the classical Laguerre polynomials is given by

$$\hat{L}_{n,m}^k(x) = L_m^k(-x)L_{n-m}^{k-1}(x) + L_m^{k-1}(-x)L_{n-m-1}^k(x); \quad \text{for } n \geq m. \quad (2.51)$$

For a specific value of m , that is, when $m = 0$, it is possible to reconstruct the classical Laguerre polynomials and weight function.

$$\hat{L}_{n,0}^k(x) = L_n^k(x), \quad (2.52)$$

$$W_0^k = W^k = x^k e^{-x}. \quad (2.53)$$

2.5 Supersymmetry in quantum mechanics

In this section, we provide a comprehensive overview of the mathematical structure of SUSY QM [1, 40] and its pivotal role. SUSY has emerged as a powerful theoretical framework that enables the systematic derivation of both bound and scattering states in ES systems. Moreover, the application of SUSY QM extends beyond the analysis of known ES systems; it also aids in the discovery of new ES systems.

Let us start with a single particle one dimensional Hamiltonian (in units of $2m = 1$, $\hbar = 1$)

$$H_1 = -\frac{d^2}{dx^2} + V_1(x). \quad (2.54)$$

Let $\psi_1^0(x)$ be the ground state wave function corresponding to this Hamiltonian then this $\psi_1^0(x)$ satisfy the Schrödinger equation

$$H_1 \psi_1^0(x) = -\frac{d^2 \psi_1^0(x)}{dx^2} + V_1(x) \psi_1^0(x) = E_1^0 \psi_1^0(x). \quad (2.55)$$

Without loss of generality, we can set the ground state energy to $E_1^0 = 0$. This simplifies the Schrödinger equation for the ground state to

$$-\frac{d^2\psi_1^0(x)}{dx^2} + V_1(x)\psi_1^0(x) = 0, \quad (2.56)$$

thus, we get

$$V_1(x) = \frac{\psi_1^{0''}(x)}{\psi_1^0(x)}. \quad (2.57)$$

Here, the prime symbol denotes differentiation with respect to x .

The Hamiltonian, as expressed in equation (2.54), can be reformulated using the ground state wavefunction

$$H_1 = -\frac{d^2}{dx^2} + \frac{\psi_1^{0''}(x)}{\psi_1^0(x)}. \quad (2.58)$$

Following the previous discussion, it is now easy to factorize the Hamiltonian in term of linear differential operator as

$$H_1 = A^\dagger A, \quad (2.59)$$

where the linear differential operators are defined as

$$A = \frac{d}{dx} + W(x) \quad \text{and} \quad A^\dagger = -\frac{d}{dx} + W(x). \quad (2.60)$$

Eq. (2.59) and (2.60), enable us to identify

$$V_1(x) = W^2(x) - W'(x). \quad (2.61)$$

This is the well-known equation called the Riccati equation. In SUSY QM, the term $W(x)$ is commonly known as the "Superpotential". The solution for superpotential $W(x)$ in terms of known ground state wave function is given by

$$W(x) = -\frac{\psi_1^{0'}(x)}{\psi_1^0(x)}. \quad (2.62)$$

This solution is achieved by realizing that once $A\psi_1^0 = 0$ is satisfied, it automatically provides a solution to $H_1\psi_1^0 = A^\dagger A\psi_1^0 = 0$. Now, to construct the SUSY theory associated with Hamiltonian H_1 , it is necessary to define another Hamiltonian H_2 by reversing the sequence of two differential operators A and $A^\dagger$

$$H_2 = AA^\dagger. \quad (2.63)$$

A bit of simplification reveals that the operator H_2 is actually a Hamiltonian associated with a new potential $V_2(x)$. i.e.

$$H_2 = AA^\dagger = -\frac{d^2}{dx^2} + V_2(x), \quad (2.64)$$

where potential V_2 is

$$V_2(x) = W^2(x) + W'(x). \quad (2.65)$$

The potentials V_1 and V_2 are known as SUSY partner potentials, and Hamiltonians H_1 and H_2 associated with these potentials are termed supersymmetric partner Hamiltonians. These two partner Hamiltonians are isospectral except for the ground state energy. Their eigenfunctions and S-matrices are also related to each other.

2.5.1 Eigenvalues and Eigenfunctions of partner Hamiltonians

Let ψ_1^n and ψ_2^n represent the eigenfunctions associated with the Hamiltonians H_1 and H_2 , respectively, each with corresponding energy eigenvalues E_1^n and E_2^n .

For $n > 0$, Schrödinger equation corresponding to the Hamiltonian H_1 is

$$H_1\psi_1^n(x) = A^\dagger A\psi_1^n(x) = E_n^{(1)}\psi_1^n(x), \quad (2.66)$$

this implies

$$H_2(A\psi_1^n) = AA^\dagger A\psi_1^n = E_n^{(1)}(A\psi_1^n). \quad (2.67)$$

Similarly Schrödinger equation for partner Hamiltonian H_2 is

$$H_2\psi_2^n = AA^\dagger\psi_2^n = E_n^{(2)}\psi_2^n. \quad (2.68)$$

Using this equation, we can express the following

$$H_1(A^\dagger\psi_2^n) = A^\dagger AA^\dagger\psi_2^n = E_n^{(2)}(A^\dagger\psi_2^n). \quad (2.69)$$

From Eqs. (2.66)-(2.69) and considering the fact that $E_0^{(1)} = 0$, it is evident that the eigenvalues and eigenfunctions of the two Hamiltonians, H_1 and H_2 , are interconnected as

$$E_2^n = E_1^{n+1}, \quad E_1^0 = 0, \quad (2.70)$$

$$\psi_2^n = [E_1^{n+1}]^{-1/2} A\psi_1^{n+1}, \quad (2.71)$$

$$\psi_1^{n+1} = [E_2^n]^{-1/2} A^\dagger\psi_2^n. \quad (2.72)$$

We can clearly see that the operator A destroys a node, while the operator $A^\dagger$ creates an additional node. This means they function as annihilation and creation operators, respectively. If the wavefunction ψ_1^{n+1} associated with Hamiltonian H_1 is normalized, then the wavefunction ψ_2^n corresponding to Hamiltonian H_2 is also normalized, and vice versa. Possessing knowledge of all the wavefunction of the original Hamiltonian H_1 enables one to determine all the wavefunctions corresponding to the SUSY partner Hamiltonian H_2 using operator A . Additionally all eigenfunctions of H_1 except ground state can be reconstructed by the wavefunction of H_2 by using the operator $A^\dagger$.

To gain a deeper understanding of the degeneracy of energy levels of H_1 and H_2 , we introduce a SUSY Hamiltonian H . This SUSY Hamiltonian H is represented as a matrix and includes both partner Hamiltonians H_1 and H_2 as

$$H = \begin{bmatrix} H_1 & 0 \\ 0 & H_2 \end{bmatrix}. \quad (2.73)$$

This matrix Hamiltonian is embedded within a closed algebra that encompasses both bosonic and fermionic operators, governed by commutation and anti-commutation relations. Consider the operators

$$Q = \begin{bmatrix} 0 & 0 \\ A & 0 \end{bmatrix} \quad (2.74)$$

and

$$Q^\dagger = \begin{bmatrix} 0 & A^\dagger \\ 0 & 0 \end{bmatrix}. \quad (2.75)$$

in conjunction with Hamiltonian H . These operators Q and $Q^\dagger$ are known as supercharges. The closed superalgebra $sl(1/1)$ described by following commutation and anticommutation relations as

$$\begin{aligned} [H, Q] &= [H, Q^\dagger] = 0, \\ \{Q, Q^\dagger\} &= H, \quad \{Q, Q\} = \{Q^\dagger, Q^\dagger\} = 0. \end{aligned} \quad (2.76)$$

The degeneracy in the system arises from the fact that the supercharges Q and $Q^\dagger$ commute with the Hamiltonian H . These supercharges can be understood as operators that convert bosonic degrees of freedom into fermionic ones and vice versa. The ground state energy is required to be zero in the case of unbroken SUSY, and non-zero when SUSY is broken.

2.5.2 Hierarchy of Hamiltonians

As discussed in the previous section, if the ground state energy of a Hamiltonian, denoted as H_1 , is zero, then it can always be represented in a factorizable form as a product of two linear differential operators. Therefore, if the ground state energy and eigenfunction of a Hamiltonian H_1 is E_1^0 and ψ_1^0 respectively, then we can write based on eq. (2.59), the following

$$H_1 = A_1^\dagger A_1 + E_1^0 = -\frac{d^2}{dx^2} + V_1(x), \quad (2.77)$$

where

$$A_1 = \frac{d}{dx} + W_1(x) \quad \text{and} \quad A_1^\dagger = -\frac{d}{dx} + W_1(x). \quad (2.78)$$

W_1 is superpotential and given by

$$W_1(x) = -\frac{\psi_1^{0'}(x)}{\psi_1^0(x)}. \quad (2.79)$$

Now, we can easily write the SUSY partner Hamiltonian H_2 as

$$H_2 = A_1 A_1^\dagger + E_1^0 = -\frac{d^2}{dx^2} + V_2(x), \quad (2.80)$$

where

$$V_2(x) = W_1^2 + W_1' + E_1^0 = V_1(x) + 2W_1' = V_1 - 2\frac{d^2}{dx^2} \ln \psi_1^0. \quad (2.81)$$

The eigenfunctions and eigenvalues of these two partner Hamiltonians are related by following

$$E_2^n = E_1^{n+1}, \quad \psi_2^n = [E_1^{n+1} - E_1^0]^{-1/2} A_1 \psi_1^{n+1}. \quad (2.82)$$

Considering the Hamiltonian H_2 with the ground state energy $E_2^0 = E_1^1$, it can be expressed as

$$H_2 = A_1 A_1^\dagger + E_1^0 = A_2^\dagger A_2 + E_1^1, \quad (2.83)$$

where

$$A_2 = \frac{d}{dx} + W_2(x), \quad A_2^\dagger = -\frac{d}{dx} + W_2(x) \quad \text{and} \quad W_2(x) = -\frac{d}{dx} \ln \psi_2^0. \quad (2.84)$$

prime denote the derivative with respect to x . Now, it is possible to construct a new Hamiltonian, denoted as H_3 , which serves as the SUSY partner to the Hamiltonian H_2 .

$$H_3 = A_2 A_2^\dagger + E_1^1 = -\frac{d^2}{dx^2} + V_3(x), \quad (2.85)$$

where

$$\begin{aligned} V_3(x) &= W_2^2 + W_2' + E_1^{(1)} = V_2(x) - 2\frac{d^2}{dx^2} \ln \psi_1^0 \\ &= V_1(x) - 2\frac{d^2}{dx^2} \ln(\psi_1^0 \psi_2^0). \end{aligned} \quad (2.86)$$

Thus, eigenvalues and eigenfunctions are

$$\begin{aligned} E_3^n &= E_2^{n+1} = E_1^{n+2}, \\ \psi_n^{(3)} &= (E_2^{n+1} - E_2^0)^{-1/2} A_2 \psi_2^{n+1} \\ &= (E_1^{n+2} - E_1^1)^{-1/2} (E_1^{n+2} - E_1^0)^{-1/2} A_2 A_1 \psi_1^{n+2}. \end{aligned} \quad (2.87)$$

Thus, continuing in this way, it becomes evident that if the original Hamiltonian H_1 possesses q (≥ 1) bound states characterized by eigenvalues E_1^n and eigenfunctions ψ_1^n , where $0 \leq n \leq (q-1)$, then it is possible to construct a sequence of $(q-1)$ Hamiltonians $H_2, H_3, \dots, H_q$. In this sequence, the r^{th} Hamiltonian (H_r) will share the identical eigenvalue spectrum with H_1 , with the exception that the first $(r-1)$ eigenvalues of H_1 are absent in H_r . For $r = 2, 3, \dots, q$, we can write

$$H_r = A_r^\dagger A_r + E_{r-1}^{(1)} = -\frac{d^2}{dx^2} + V_r(x), \quad (2.88)$$

where,

$$A_r = \frac{d}{dx} + W_r(x), \quad W_r(x) = -\frac{d}{dx} \ln \psi_r^0. \quad (2.89)$$

One also has

$$\begin{aligned} E_r^{(n)} &= E_{r-1}^{n+1} = \dots = E_1^{n+r-1}, \\ \psi_r^{(n)} &= (E_1^{n+r-1} - E_1^{r-2})^{-1/2} \dots (E_1^{n+r-1} - E_1^0)^{-1/2} A_{r-1} \dots A_1 \psi_1^{n+r-1}, \\ V_r(x) &= V_1(x) - 2\frac{d^2}{dx^2} \ln(\psi_1^0 \dots \psi_{r-1}^0). \end{aligned} \quad (2.90)$$

Thus, by understanding all the eigenvalues and eigenfunctions of H_1 , we can instantly determine the energy eigenvalues and eigenfunctions for the entire hierarchy of $q-1$ Hamiltonians.

2.6 Shape Invariance

In 1983, Gendenshtein [18] introduced the concept of a SIPs within the framework of SUSY QM. A pair of SUSY partner potentials $V_1(x)$ and $V_2(x)$ are considered to be SI if they have

similar forms and differ only in the parameters that appear in them. In a more precise manner, if the partner potentials, $V_1(x; a_1)$ and $V_2(x; a_2)$ satisfy the condition

$$V_2(x; a_1) = V_1(x; a_2) + R(a_1), \quad (2.91)$$

where a_1 represents a set of parameters, a_2 is defined as a function of a_1 (let $a_2 = f(a_1)$) and the remainder $R(a_1)$ is independent of x , then potentials, $V_1(x; a_1)$ and $V_2(x; a_1)$ are said to be SI. Utilizing the condition in Eq. (2.91) and the hierarchy of Hamiltonians discussed in the previous section, one can readily determine the energy eigenvalues and eigenfunctions for any SIPs in the case of unbroken SUSY. To achieve this, we consider the case of unbroken SUSY, for which

$$E_1^0(a_1) = 0, \quad \psi_1^0(x; a_1) = N \exp \left(- \int^x W_1(y; a_1) dy \right). \quad (2.92)$$

The next step is to construct a family of Hamiltonians H_p for $p = 1, 2, 3, \dots$. From the discussion in the previous section, it becomes evident that if Hamiltonian H_1 possesses q bound states, then it is possible to construct $q - 1$ Hamiltonians $H_1, H_2, \dots, H_{q-1}$. The n^{th} Hamiltonian H_n will share the same spectrum as H_1 , with the exception that the first $n - 1$ levels present in H_1 will not appear in H_n . By repeatedly applying the SI condition, as referred to in Eq. (2.91), Hamiltonian H_p can be written as

$$H_p = -\frac{d^2}{dx^2} + V_1(x; a_p) + \sum_{k=1}^{p-1} R(a_k), \quad (2.93)$$

where $a_p = f^{p-1}(a_1)$, meaning the function f applied $p - 1$ times. Similarly, we can construct the Hamiltonian H_{p+1} using equations (2.91) and (2.93)

$$\begin{aligned} H_{p+1} &= -\frac{d^2}{dx^2} + V_1(x; a_{p+1}) + \sum_{k=1}^p R(a_k) \\ &= -\frac{d^2}{dx^2} + V_2(x; a_p) + \sum_{k=1}^{p-1} R(a_k). \end{aligned} \quad (2.94)$$

From the preceding discussion, it is understood that H_p and H_{p+1} are SUSY partner Hamiltonians and therefore, they share identical bound state spectra, except for the ground state of H_p , whose energy is

$$E_p^0 = \sum_{k=1}^{p-1} R(a_k). \quad (2.95)$$

Reversing the sequence from H_p to H_{p-1} and so on, we ultimately arrive at H_2 and H_1 , where the ground state energy E_1^0 is zero and the n^{th} level coincides with the ground state

of the Hamiltonian H_n . Consequently, the n^{th} eigenvalue of H_1 is given by

$$E_1^n(a_1) = \sum_{k=1}^n R(a_k); \quad E_1^0(a_1) = 0. \quad (2.96)$$

and eigenfunction is given by

$$\psi_1^n(x; a_1) \propto A^\dagger(x; a_1) A^\dagger(x; a_2) \dots A^\dagger(x; a_n) \psi_1^0(x; a_{n+1}). \quad (2.97)$$

2.7 Point Canonical Transformation Approach

The PCT approach is a very effective method for constructing new SIPs and non-SIPs. This tool was first used by Quesne [131] to generate the new ES SI systems. This approach is applicable in standard context [240, 241] along with QES [242], CES [243], and position-dependent effective mass (PDEM) systems [217]. In this thesis, PCT approach is used for PDEM systems. Here we briefly mention important mathematical steps of PCT approach.

We start with Schrödinger equation ($\hbar = 2m = 1$)

$$\frac{d^2\psi(x)}{dx^2} + \left(E_n - V(x)\right)\psi(x) = 0. \quad (2.98)$$

In PCT approach, to solve the above Schrödinger equation, one assumes the solution of the following form [231, 244, 245]

$$\psi(x) = f(x)F(g(x)), \quad (2.99)$$

where $f(x)$, $g(x)$ are two as yet undetermined functions and $F(g)$ will subsequently be identified as one of the orthogonal polynomials, it satisfies a second-order differential equation

$$\frac{d^2F}{dg^2} + Q(g)\frac{dF}{dg} + R(g)F = 0. \quad (2.100)$$

Now substituting $\psi(x)$ from Eq. (2.99) in (2.98), it gives

$$\frac{d^2F}{dg^2} + \left(\frac{g''}{g'^2} + \frac{2f'}{fg'}\right)\frac{dF}{dg} + \left(\frac{f''}{fg'^2} + (E - V_{eff})\frac{1}{g'^2}\right)F = 0, \quad (2.101)$$

and comparing with Eq. (2.100), one obtains

$$Q(g(x)) = \left(\frac{g''}{g'^2} + \frac{2f'}{fg'}\right), \quad (2.102)$$

$$R(g(x)) = \left(\frac{f''}{f g'^2} + (E - V_{eff}) \frac{1}{g'^2} \right). \quad (2.103)$$

Eq. (2.102) allows one to find $f(x)$, which is given by [244]

$$f(x) \propto \frac{1}{\sqrt{g'}} \exp \left(\frac{1}{2} \int^{g(x)} Q(t) dt \right), \quad (2.104)$$

Putting it in (2.103), one can obtain

$$E - V(x) = \frac{1}{2} \{g(x), x\} + g'^2 \left(R - \frac{1}{2} Q' - \frac{1}{4} Q^2 \right). \quad (2.105)$$

where $\{g(x), x\}$ is the Schwartzian derivative given by

$$\{g(x), x\} = \frac{g'''}{g'} - \frac{3}{2} \left(\frac{g''}{g'} \right)^2. \quad (2.106)$$

In the left hand side of Eq. (2.105), an energy term E is present, to compensate this we ensure that an energy like term must be present in right hand side of this equation. For this purpose one can choose an appropriate form of $g(x)$. The x dependent term give rise to a potential $V(x)$ with well behaved solution. By using Eqs. (2.99) and (2.104) one can obtain the wavefunction in PCT approach

$$\psi(x) = N \times (g'(x))^{-\frac{1}{2}} \exp \left(\frac{1}{2} \int Q(g) dg \right) F(g(x)). \quad (2.107)$$

If the function $F(g)$ is equivalent to one of the EOPs, then the corresponding potentials obtained from Eq. (2.105) will provide new rationally extended potentials.

2.8 Group Theoretic Approach

In this thesis, group theoretic approach is used to generate QES potential. The associated group is known as potential group $SO(2, 1)$ [118–120]. A potential group links states with identical energy levels but different potential strength. In this section, we provide a brief overview of this approach. The differential realization of generators of $SO(2, 1)$ group is given by

$$J_0 = -i \frac{\partial}{\partial \phi}, \quad J_{\pm} = e^{\pm i \phi} \left[\frac{\partial}{\partial x} + F(x) \left(i \frac{\partial}{\partial \phi} \mp \frac{1}{2} \right) + G(x) \right], \quad (2.108)$$

which govern the underlying commutation relations of $SO(2, 1)$ as

$$[J_+, J_-] = -2J_0, \quad [J_0, J_{\pm}] = \pm J_{\pm} \quad (2.109)$$

where, $F(x)$ and $G(x)$ are real functions, and the generators J_+ and J_- of $SO(2, 1)$ potential algebra are Hermitian conjugates of each other. These generators fulfill the commutation relation referred to in (2.109). Consequently, we obtain two coupled equations.

$$\frac{dF}{dx} = 1 - F^2, \quad \text{and} \quad \frac{dG}{dx} = -FG, \quad (2.110)$$

In this thesis, the discussion is confined to bound states only, for which one can consider the unitary irreducible representation of the D_k^+ -type. This representation of the $so(2, 1)$ potential algebra has basis states that serve as eigenstates for different Hamiltonians but have the same energy level. The Casimir operator for the $SO(2, 1)$ algebra in terms of the above generators is given by

$$J^2 = J_0^2 \mp J_0 - J_{\pm}J_{\mp}. \quad (2.111)$$

it can be explicitly expressed by

$$J^2 = (\partial^2/\partial x^2) - F' \left[(\partial^2/\partial \phi^2) + \frac{1}{4} \right] + 2iG' (\partial/\partial \phi) - G^2 - \frac{1}{4}. \quad (2.112)$$

In the realization of (2.108), with the basis state, generators follow the relations

$$J_0|km\rangle = m|km\rangle, \quad (2.113)$$

$$J^2|km\rangle = k(k-1)|km\rangle, \quad \text{with } (m = k, k+1, k+2, \dots) \quad (2.114)$$

where, the given states $|km\rangle = \psi_{km}(x)e^{im\phi}$ form the basis, it follows that the coefficient functions $\psi_{km}(x)$ satisfy the Schrödinger equation

$$-\psi_{km}'' + V_m\psi_{km} = -\left(k - \frac{1}{2}\right)^2 \psi_{km}, \quad (2.115)$$

where V_m is a one-parameter family of potentials and expressed in terms of the guiding functions $F(x)$ and $G(x)$

$$V_m = \left(\frac{1}{4} - m^2\right) F' + 2mG' + G^2. \quad (2.116)$$

Wu and Alhassid [118, 129, 246], choosen a particular solutions to Eq. (2.110) that correspond to Morse, Pöschl-Teller, and Rosen-Morse potentials. Later, Englefield and Quesne [247] extended this approach, identifying more general solutions based on the conditions $F^2 < 1$, $F^2 = 1$ or $F^2 > 1$ namely,

$$(I) \quad F(x) = \tanh(x), \quad G(x) = b \operatorname{sech}(x), \quad (2.117)$$

$$(II) \quad F(x) = \pm 1, \quad G(x) = b e^{\mp x}, \quad (2.118)$$

$$(III) \quad F(x) = \coth(x), \quad G(x) = b \operatorname{cosech}(x). \quad (2.119)$$

After substituting these solutions into Eq. (2.116), one obtains the non-singular Gendenshtein, Morse, and singular Gendenshtein potentials. These solutions include those derived by Wu and Alhassid, with the Gendenshtein potentials being disguised forms of the Pöschl-Teller potentials. Additionally, for specific parameter values, the singular Gendenshtein potentials reduce to the non-singular Rosen-Morse potentials.

After laying down these mathematical preliminaries, we are now at the position to proceed for the new works presented in this thesis.